



Customer Shopping Behavior Analysis

End-to-End Data Analytics Project using Python, PostgreSQL, SQL & Power BI

Project Overview

Customer behavior plays a critical role in helping retail businesses improve customer satisfaction, increase revenue, and optimize marketing strategies. Understanding purchasing patterns allows businesses to make informed decisions about product offerings, customer segmentation, and promotional campaigns.

This project analyzes customer shopping behavior using transactional retail data containing **3,900 customer purchase records**. The complete analytics pipeline includes data cleaning in Python, data storage and querying in PostgreSQL, business analysis using SQL, and interactive dashboard creation in Power BI.

The objective was to transform raw customer transaction data into actionable business insights that support strategic business decision-making.

Business Problem

Retail organizations collect large amounts of customer transaction data every day. However, raw data alone cannot provide meaningful insights without proper analysis. The key business challenge is to identify purchasing trends, customer preferences, subscription behavior, revenue patterns, and product performance so that management can make data-driven decisions.

This project aims to answer these questions by analyzing customer shopping behavior and presenting the findings through an interactive Power BI dashboard.

Project Objectives

The primary objectives of this project are:

- Analyze customer purchasing behavior across different demographics.
- Identify revenue trends across product categories.
- Measure customer spending patterns.
- Evaluate subscription and loyalty behavior.
- Analyze customer satisfaction using review ratings.
- Build an interactive Power BI dashboard for business stakeholders.
- Generate actionable business recommendations.

Dataset Overview

Attribute	Details
Dataset Name	Customer Shopping Behavior Dataset
Business Domain	Retail Analytics
Total Records	3,900 Customer Transactions
Total Features	18 Columns
Primary Identifier	Customer ID
Missing Values	37 (Review Rating) – Imputed using Mean
Duplicate Records	None Found
Data Source	Customer Shopping Behavior Dataset (CSV)
Database Used	PostgreSQL
Final Output	Interactive Power BI Dashboard with Business Insights

Dataset Features

The dataset contains customer information including:

- Customer ID
- Age
- Gender
- Item Purchased
- Category
- Purchase Amount

- Location
- Size
- Color
- Season
- Review Rating
- Subscription Status
- Shipping Type
- Discount Applied
- Promo Code Used
- Previous Purchases
- Payment Method
- Frequency of Purchases

Data Cleaning & Preprocessing

The raw dataset was cleaned and prepared using Python before loading into PostgreSQL.

The preprocessing included:

1.) Missing Value Handling

The **Review Rating** column contained missing values.

These missing values were replaced using the average review rating to preserve data consistency without removing records.

2.) Duplicate Check

The dataset was examined for duplicate records to ensure data integrity.

3.) Data Validation

Column names, data types, and categorical values were validated before importing into PostgreSQL.

4.) Feature Engineering

Additional business-friendly calculations and categories were prepared for easier SQL analysis and Power BI visualization.

PostgreSQL Database

After preprocessing, the cleaned dataset was imported into PostgreSQL.

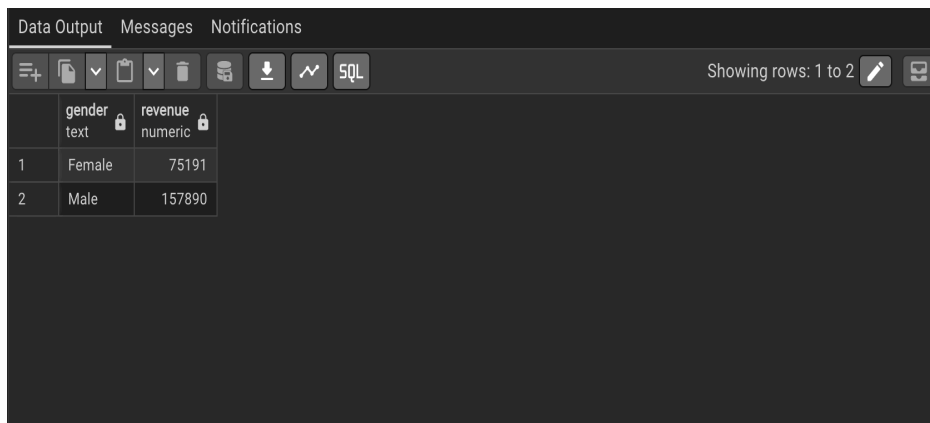
PostgreSQL served as the central database for performing business analysis using SQL.

The database enabled efficient querying of customer transactions and aggregation of business metrics.

SQL Business Analysis

SQL was used to answer important business questions including:

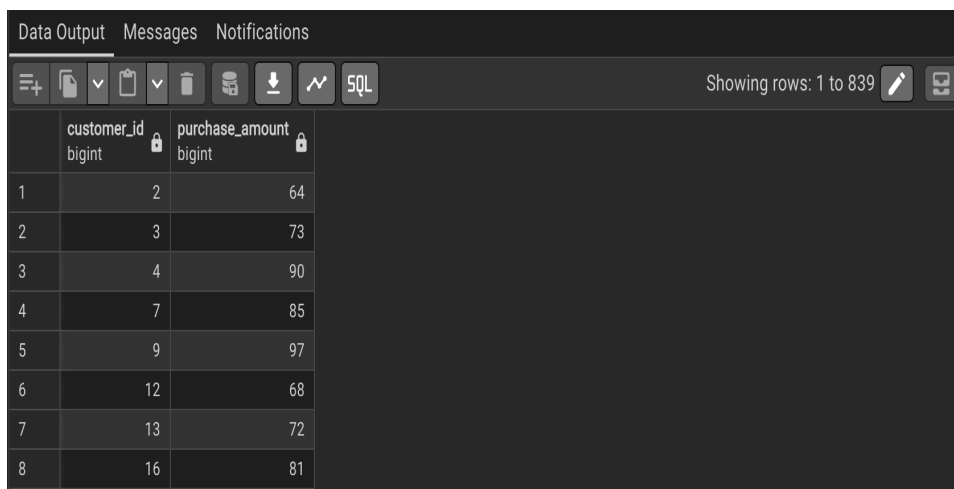
- Total revenue by gender



The screenshot shows a database interface with a dark theme. At the top, there are tabs for 'Data Output', 'Messages', and 'Notifications'. Below the tabs is a toolbar with various icons, including a plus sign, a document, a dropdown, a trash, a database icon, a download, a refresh, and an 'SQL' button. On the right side of the toolbar, it says 'Showing rows: 1 to 2'. The main area displays a table with two columns: 'gender' (text) and 'revenue' (numeric). The table has two rows: Row 1 shows 'Female' with a revenue of 75191, and Row 2 shows 'Male' with a revenue of 157890.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

- High-Spending Discount Users – Identified customers who used discounts but still spent above the average purchase amount.



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	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81

- Top 5 Products by Rating – Found products with the highest average review ratings.

Data Output Messages Notifications		
	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

- Shipping Type Comparison – Compared average purchase amounts between Standard and Express shipping.

Data Output Messages Notifications		
	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

- Subscribers vs. Non-Subscribers – Compared average spend and total revenue across subscription status.

Data Output Messages Notifications				
	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

- Discount-Dependent Products – Identified 5 products with the highest percentage of discounted purchases.

Data Output Messages Notifications		
	item_purchased text	discount_rate numeric
1	Hat	5000.00
2	Sneakers	4900.00
3	Coat	4900.00
4	Sweater	4800.00
5	Pants	4700.00

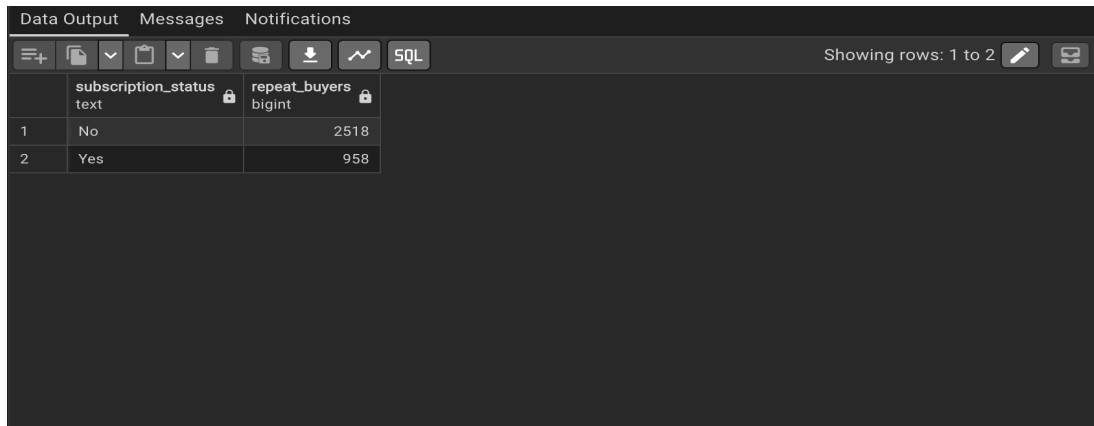
- Customer Segmentation – Classified customers into New, Returning, and Loyal segments based on purchase history.

Data Output Messages Notifications		
	customer_segment text	Number of customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

- Top 3 Products per Category – Listed the most purchased products within each category.

Data Output Messages Notifications				
	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

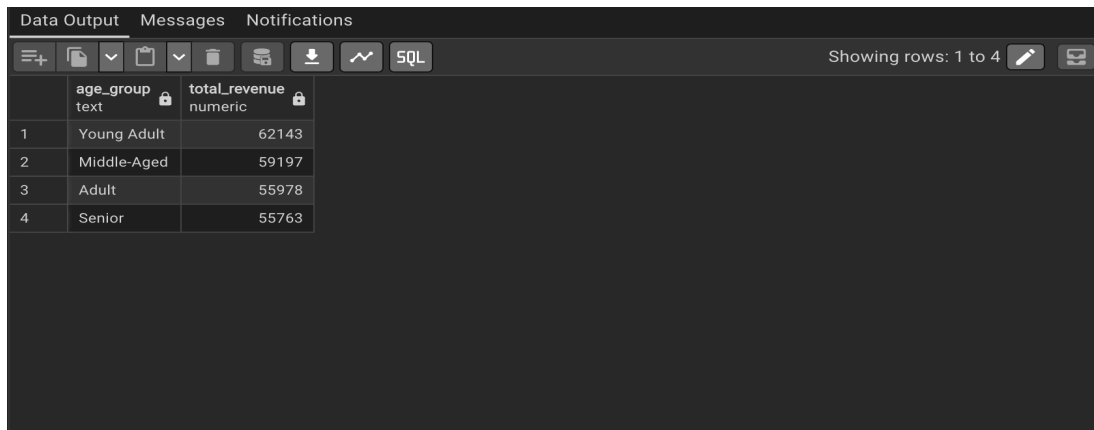
- Repeat Buyers & Subscriptions – Checked whether customers with >5 purchases are more likely to subscribe.



The screenshot shows a data table with two columns: 'subscription_status' (text) and 'repeat_buyers' (bigint). The table has two rows: Row 1 shows 'No' for subscription status and 2518 for repeat buyers. Row 2 shows 'Yes' for subscription status and 958 for repeat buyers. The interface includes tabs for 'Data Output', 'Messages', and 'Notifications', and a toolbar with icons for file operations and a 'SQL' button. The status bar indicates 'Showing rows: 1 to 2'.

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

- Revenue by Age Group – Calculated total revenue contribution of each age group.



The screenshot shows a data table with two columns: 'age_group' (text) and 'total_revenue' (numeric). The table has four rows: Row 1 shows 'Young Adult' for age group and 62143 for total revenue. Row 2 shows 'Middle-Aged' for age group and 59197 for total revenue. Row 3 shows 'Adult' for age group and 55978 for total revenue. Row 4 shows 'Senior' for age group and 55763 for total revenue. The interface includes tabs for 'Data Output', 'Messages', and 'Notifications', and a toolbar with icons for file operations and a 'SQL' button. The status bar indicates 'Showing rows: 1 to 4'.

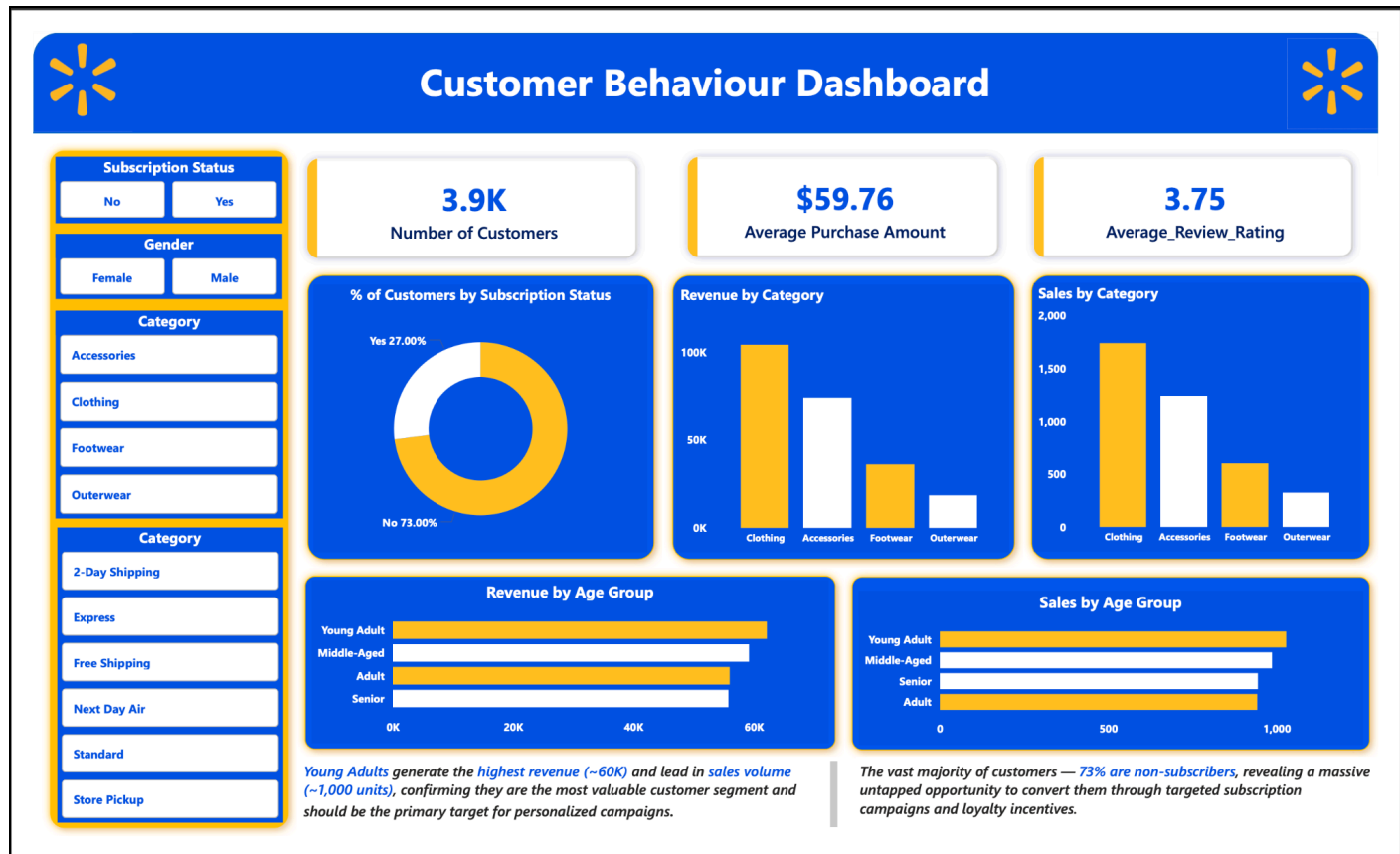
	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-Aged	59197
3	Adult	55978
4	Senior	55763

The analysis involved SQL concepts such as:

- SELECT
- WHERE
- GROUP BY
- ORDER BY
- HAVING
- Aggregate Functions
- CASE Statements
- Joins
- Subqueries

Dashboard Development

An interactive Power BI dashboard was developed to present business insights visually. The dashboard enables stakeholders to explore customer purchasing behavior dynamically.



The dashboard includes:

KPI Cards

- Total Customers
- Average Purchase Amount
- Average Review Rating

Interactive Filters

- Subscription Status
- Gender
- Product Category

- Shipping Method

Visualizations

- Revenue by Category
- Sales by Category
- Revenue by Age Group
- Sales by Age Group
- Subscription Distribution

The dashboard enables stakeholders to explore customer purchasing behavior dynamically.

Key Business Insights

The analysis revealed several important findings:

1.) Young Adults generated the highest revenue.

This customer segment contributes the largest share of overall revenue and should be considered the highest-value customer group.

2.) Clothing is the highest-performing product category.

It leads both revenue generation and sales volume compared to other categories.

3.) Subscription penetration remains low.

Approximately **73% of customers are non-subscribers**, indicating significant opportunities to increase customer loyalty through subscription programs.

4.) Average customer spending is consistent.

Customers spend an average of **\$59.76 per purchase**, showing relatively stable purchasing behavior.

5.) Customer satisfaction is positive.

The average review rating is **3.75 out of 5**, suggesting generally positive customer experiences while highlighting opportunities for quality improvement.

Business Recommendations

Based on the analysis, the following recommendations are proposed:

- Launch targeted subscription campaigns for non-subscribers.
- Focus marketing efforts on Young Adult customers.
- Increase investment in high-performing product categories, especially Clothing.
- Implement personalized marketing using customer purchase history.
- Monitor lower-rated products to improve customer satisfaction.
- Optimize promotional campaigns based on customer purchasing patterns.

Analytics Workflow

Phase	Activity Performed	Tool Used	Output
1. Data Collection	Imported raw customer transaction dataset	CSV	Raw Dataset
2. Data Cleaning	Removed inconsistencies, handled missing values, validated data types	Python (Pandas)	Clean Dataset
3. Feature Engineering	Prepared data for business analysis and reporting	Python	Analysis-Ready Dataset
4. Database Integration	Imported cleaned dataset into PostgreSQL	PostgreSQL	Relational Database
5. SQL Analysis	Solved business questions using SQL queries	SQL	Business Insights

6. Dashboard Development	Built interactive dashboard with KPIs and visualizations	Power BI	Executive Dashboard
7. Business Recommendations	Derived actionable insights from the analysis	Power BI + SQL	Strategic Recommendations

Tools & Technologies

Tools / Technology	Purpose
Python	Data cleaning, preprocessing, missing value treatment, and feature engineering
Pandas	Data manipulation and preprocessing
NumPy	Numerical computations and data transformation
PostgreSQL	Relational database for storing cleaned data
SQL	Business analysis and answering analytical questions
Power BI	Interactive dashboard creation and visualization
VS Code	Python and SQL development environment
Jupyter Notebook	Exploratory Data Analysis (EDA) and preprocessing
Git & GitHub	Version control and project portfolio management

Project Outcome

This project successfully demonstrates a complete end-to-end data analytics workflow.

Starting from raw retail transaction data, the project performs data cleaning, database integration, SQL-based business analysis, and interactive dashboard development to generate meaningful business insights.

The findings can help organizations improve customer engagement, optimize marketing strategies, enhance product performance, and support data-driven decision-making.